NIKHIL KUMAR TIWARI

Full-Stack Developer | Computer Science (AI/ML) Student

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LinkedIn | Portfolio | GitHub

Professional Summary

Computer Science (AI/ML) student and Full-Stack Developer with experience building web applications using React, Next.js, Node.js, and Express.js. Hands-on experience with Python, Retrieval-Augmented Generation (RAG), vector embeddings, and LLM-powered applications. Interested in developing scalable web applications and integrating AI capabilities into practical software solutions.

Projects

Glucose Twin – AI-Powered Predictive Digital Twin GitHub | Demo

Next.js, React, Python, FastAPI, XGBoost, Pandas, CGMacros

- Built a healthcare dashboard to visualize glucose readings, model predictions, patient information, and risk indicators through interactive charts and a 3D digital twin interface.
- Worked on time-series data processing and predictive analytics for glucose forecasting and high/low glucose risk estimation using the project's trained models.
- Implemented replay-based simulation using historical CGMacros observations to demonstrate live monitoring while preserving original measurements and timestamps.

LeetCode Clone GitHub | Demo

Next.js, PostgreSQL, Prisma, Docker, Judge0, Better Auth

- Built a full-stack coding platform supporting multi-language code compilation, sandboxed execution, and submission verdicts through Judge0 integration.
- Implemented authentication, session management, and protected routes using Better Auth.
- Developed problem management, test-case validation, and submission history using PostgreSQL and Prisma ORM.

VisitorPass Management System GitHub | Live Demo

MongoDB, Express.js, React.js, Node.js

- Developed a visitor management application with request submission, multi-step approval workflows, and QR-code-based entry verification.
- Built REST APIs for admin, host, and visitor workflows and integrated automated email notifications to simplify approval communication.

Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python
Frontend: React.js, Next.js, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, RESTful APIs, Microservices
Databases: PostgreSQL, MongoDB, Prisma ORM, Redis
AI / Machine Learning: Scikit-learn, PyTorch, RAG, LLMs, Vector Embeddings, LangChain
Tools Git, GitHub, Docker
Core Concepts: OOP, DBMS, Operating Systems, Computer Networks, System Design Fundamentals

Education

VIT Bhopal University Bhopal, India
Bachelor of Technology in Computer Science and Engineering (AI/ML) Expected Aug. 2029
CGPA: 8.37/10

Army Public School Jodhpur, India
Intermediate (Class XII) Aug. 2024
Final Grade: 83.4%

Certifications

- Tutedude – MERN Stack Development Course
- Complete React and Next.js Course with AI-Powered Projects